

Notes from Network+ studies (foundational networking)

Note: The Network+ exam objectives are used to organize each section. Only information relevant to this lab iteration is included; unrelated objectives and concepts are omitted.

1.7 Subnetting

Strap in! (:

What is subnetting? Subnetting is the process of dividing one larger IP network into smaller networks called subnets. Its purpose is to organize networks more efficiently, reduce unnecessary broadcast traffic, and use IP addresses more cleanly. For example, instead of one large network for an entire building, a company might create separate subnets for HR, sales, and servers.

Let's talk about IPv4 addresses:

IPv4 Addresses are comprised of 4 octets – each octet (8 bits) is binary although we commonly see it translated into decimal. This means each octet can have a value from 0-255.

Each octet represents something in an IP address. The specific octets that need to be considered to identify the network location is determined by the subnet mask or CIDR notation. Basically, the subnet mask tells your computer which part of the IP address is the network identifier and which is the device identifier (Don't worry we'll get into that in a bit).

Subnet masks are picked usually determined by the amount of devices needing to connect to a local network (this can be represented with CIDR notation /# or subnet masks ###.###.###.###):

- /24 - small networks (254 devices or less)
- /16 - medium/large networks (65,534 usable host addresses)
- /8 - large networks (16,777,214 usable host addresses)

The /# refers to the number of bits used to determine the network address, so /24 mean first 3 octets because the first 3 octets are comprised of 24 bits.

In an IPv4 address with the /24 CIDR notation, the first 3 octets are typically there to indicate where the network is, the 4th octet points to which device it is on that specific network.

(e.g., 192.168.1.10, First 3 octets? Where the network is. Last octet? Which device on that network.

Take a look at this:

192.168.1.10 /24 - Printer

192.168.1.15 /24 - PC

192.168.1.1 /24 - Router

(All within the same network because the first 3 octets are all the same – that last octet? Just determines the specific device within that subnet.)

Let's rewind...

We can identify what network a device is part of depending on the CIDR notation (/8, /16, etc.). A **CIDR notation** basically tells us which part of the IPv4 address is the network portion and which is the host portion.

So working with a /24 CIDR notation:

- Network.network.network.host
- 192.168.1.10
- 192.168.1.25
- 192.168.1.200

All of these belong to the same network because the network portion of their IP address is the same.

Working with a /16 CIDR notation - we know the first 16 bits are the network portion, and the last 16 bits is the host portion.

- So 172.20.125.122, the 172.20 is the network portion (what network am I on?)
- The 125.122 is the host portion (who am I on this network?).

There are 4 main addresses to be aware of in any given network/subnet:

- Network address
- Broadcast address

- First usable host address
- Last usable host address

The **network address** is found by setting all host bits to 0. This is basically the first IP address within a network. Eg. 192.168.10.1 /16, the network address is 192.168.0.0.

The **broadcast address** is found by setting all host bits to 1. This is the last IP address available in a given network. Eg. 192.168.10.1 /16, the broadcast address is 192.168.255.255.

The **first usable host address** is found by adding one number to the network address. Eg. We've already determined that in 192.168.10.1 /16, the network address is 192.168.0.0, therefore the first usable host address will be 192.168.0.1.

The **last usable host address** is found by subtracting one number from the broadcast address. Eg. We've already determined that in 192.168.10.1 /16, the broadcast address is 192.168.255.255, therefore the last usable host address is 192.168.255.254.

Additional example:

175.222.35.24/8

Network Address - 175.0.0.0

Broadcast Address - 175.255.255.255

1st usable host address - 175.0.0.1

Last usable host address - 175.255.255.254

In practice, /8, /16, /24, are commonly treated as base (unsplit) networks because they align with the octet boundaries. Subnetting occurs when we extend beyond those boundaries by borrowing bits from the host portion to create smaller networks (e.g., /10, /18, /29).

In subnetting, understanding the binary math behind IPv4 addresses becomes vitally important. Remember each octet is 8 binary values:

128 - 64 - 32 - 16 - 8 - 4 - 2 - 1

With this in mind we can calculate:

11010000 = 208

254 = 11111110

00110011 = 51

135 = 10000111

This is crucial in determining how many subnets (mini networks) there are, and what host addresses fall within each one.

Let's look at 192.168.10.45 /26, we know that the nearest octet border is /24 so /26 means we are **borrowing** 2 bits from the host portion to create smaller networks.

The amount of subnets created can be measured by using the power of 2. In the above example (/26) we borrowed 2 bits from the host portion, so 2 to the power of 2 equals 4. So we've created 4 subnets.

- **$2^{\text{borrowedbits}}$ = number of subnets**

The key now is identifying the network address, broadcast address, and usable host range of the newly created subnets.

Pause, take a breath (remember):

192.168.1.1 /24

Network address: 192.168.1.0

Broadcast address: 192.168.1.255

Usable host range: 192.168.1.1-254

So in our subnetted example of 192.168.10.45 /26, we now have 4 separate networks we need to determine the network addresses, broadcast addresses, and usable host ranges for.

Since we have subnetted our original /24 network into 4 pieces (represented by /26), our host portion is now divided into 4 segments.

Consider this, in our /24 network, the available address range is 0-255, the same is true in /26 except now we need to map what host addresses belong to which subnet.

- We'll take 256 and divide it into 4 parts, $256/4 = 64$.
- The network address for each subnet is 0, 64, 128, 192.
- The broadcast address for each subnet is 63, 127, 191, 255.
- **So each subnet has 62 available host addresses, because each subnet total address availability is 64 (256 split in 4), and the first/last address is used for network/broadcast.**

Now let's identify which subnet certain devices are in. 192.168.10.X /26 is split into 4 subnets.

- (192.168.10.45 /26) This device is located within the first subnet because the host range falls between 192.168.10.1-62, therefore this device belongs to the 192.168.10.0 subnet.
- (192.168.10.188) This device is located within the third subnet because the host range falls between 192.168.10.129-190, therefore this device belongs to the 192.168.10.128 subnet.

Use this frame to create your own IPv4 addresses with CIDR notations/subnet masks and practice finding the network/broadcast address, and the usable host range.

I heavily focused on CIDR notations, but understand that subnet masks will tell you the same thing that a CIDR notation is, just in a different format.

Subnet mask = CIDR notation

255.0.0.0 = /8

255.255.0.0 = /16

255.255.255.0 = /24

Remember that when subnetting we're borrowing host bits to create said subnets, CIDR notations represents that borrowed bit by just adding to the /#. Subnet mask notations do it a 'bit' differently, get it?...

Anyway, borrowing a bit will be represented differently with subnets masks, so with a /8 CIDR notation is represented as 255.0.0.0, because the network portion has all 8 bits in that first octet set to 1 (binary math – take a refresher if you need to).

11111111.00000000.00000000.00000000 = 255.0.0.0

10000000 = 128 (one borrowed bit)

11000000 = 192 (two borrowed bits)

11100000 = 224 (three borrowed bits)

11110000 = 240 (four borrowed bits)

11111000 = 248 (five borrowed bits)

11111100 = 252 (six borrowed bits)

11111111 = 254 (seven borrowed bits)

11111111.11111111.11100000.00000000 = /19 (CIDR) = 255.255.224.0 (subnet mask)

So how we would represent a /17 CIDR notation in regular subnet mask format?

255.255.128.0

How about /29?

255.255.248.0

Let's do it in reverse, what's 255.240.0.0 in CIDR notation?

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Variable Length Subnet Mask (VLSM): VLSM is a subnetting method that lets you use different subnet masks within the same network. Its purpose is to avoid wasting IP addresses by giving each subnet only the size it actually needs. For example, a small department might use a /28 while a larger department uses a /24.

2.1 Routing protocols and selection

Static routing: Static routing is when a network administrator manually tells a router exactly where to send traffic for a given destination. This provides simple, and predictable path control of network traffic, however, static routes do not automatically adapt if the network changes – so if a link fails, the route usually remains untraversable.

Dynamic routing: Dynamic routing is when routers automatically learn routes from each other using routing protocols instead of relying only on manual entries. Dynamic routing is more complex and uses additional router resources, but it provides redundancy and is less prone to error. So if a path between two routers go down, a dynamic routing protocol can often find and use another path automatically.

The route a router decides to send its packet is based on a number of factors, so what exactly determines the **route selection**? **Administrative distance** is a factor to consider, it is how much the router actually trusts the provided route – a directly connected network ranks highest in trust because the router literally has an interface on that network. Static routing is also very trusted because an admin has to manually configure it. Dynamic protocols are trusted based on their default administrative distance values

(Lowest value = Most trusted: Connected route/0, Static/1, EIGRP/90, OSPF/110, RIP/120).

Prefix length is another route selection factor; routers prefer the most specific matching route when sending packets. A longer prefix length means more bits identify the network portion of an IPv4 address – so for example, a packet sent to 10.1.5.25 has three routes to choose from 10.0.0.0/8, 10.1.0.0/16, and 10.1.5.0/24 (the most specific path would be 10.1.5.0/24, it has the largest prefix length meaning it's the most specific route). Finally, a **metric** is a routing protocol's measurement of how good or bad a path is. If a router has multiple routes to the same destination from the same protocol, it chooses the route with the best metric (lowest metric) - For example, RIP uses hop count (so the lowest hop count is preferred), OSPF uses cost, commonly based on bandwidth (so the lowest cost is preferred), EIGRP uses factors such as bandwidth and delay, generally preferring paths with higher bandwidth and lower delay.

Remember as a general rule:

Highest prefix length wins first, because the most specific matching route is preferred.

Lowest administrative distance wins next, because the most trusted route source is preferred.

Lowest metric wins last, when comparing routes from the same routing protocol/source.

5.3 Troubleshooting network services

Route selection (routing table): Route selection is the process a router or Layer 3 device uses to choose where to send traffic based on its routing table. The device compares the destination IP address to its known routes and usually chooses the most specific matching route, called the longest prefix match. If multiple routes match equally, the device may then consider administrative distance and metric.

Route selection (default routes): A default route is the fallback route used when a device does not have a more specific route to the destination network. In IPv4, it is written as 0.0.0.0/0; in IPv6, it is written as ::/0. Default routes are commonly used to send unknown or internet-bound traffic toward an upstream router, firewall, or ISP.

Address pool exhaustion: Address pool exhaustion occurs when a DHCP server runs out of available IP addresses in a scope. This can happen when too many clients are on the network, the DHCP range is too small, lease times are too long, old leases are not clearing quickly enough, or unauthorized devices are consuming addresses. When the pool is exhausted, new clients may fail to receive an IP address or may assign themselves an APIPA address.

Incorrect default gateway: Incorrect default gateway means a device has the wrong router/gateway IP configured for its subnet. The device may still communicate with devices on its local subnet, because local traffic does not need the default gateway. But when it tries to reach another subnet or the internet, traffic fails because the device sends non-local traffic to the wrong gateway.

Incorrect IP address: Incorrect IP address means a device has an IP address that does not belong to the correct network/subnet, conflicts with another device, or was manually configured incorrectly. The device may appear connected physically, but it may not communicate properly because its IP address does not match the network it is actually on.

Duplicate IP address: Duplicate IP address means two devices on the same network are using the same IP address. This causes an IP conflict because the network cannot reliably know which device should receive traffic for that IP. One or both devices may lose connectivity, have intermittent access, show IP conflict warnings, or behave unpredictably.

Incorrect subnet mask: Incorrect subnet mask occurs when a device is configured with the wrong subnet mask/prefix length for its network. This causes the device to calculate the wrong local subnet, which can break communication by making the device treat remote addresses as local or local addresses as remote.